Advanced Programming Practices
MTech 2026
Lab - 12

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP12_Qn.c (e.g. ABC2026001_APP12_Q1.c).
- Every program must compile without warnings using the gcc compiler.
- Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
- Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
- Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.

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1. Write a C program to represent a general tree and perform the following operations:
    a. Find the parent of a given node.
    b. Display all children of a given node.
    c. Count the total number of leaf nodes in the tree.

2. For a given binary tree, write a program to count the total number of nodes, internal nodes and external nodes.

3. Write a C program to find the k-th largest element in an array using a heap.

Assumptions:

1. The array contains n integers.
2. The value of k satisfies $1 \leq k \leq n$.
3. Duplicate elements are allowed and are counted as separate positions in descending sorted order.

Example:

Array: 10, 5, 20, 20, 8

$k = 2$

Output: 20